Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 October 2026
Team ID	NA
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	4 Marks

Functional Requirements:
Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Use the cleaned EV dataset ev_market_clean.csv (199,999 records, 37 attributes including region, category, battery type, battery capacity, efficiency, charging time, motor power, drivetrain, price, insurance and sales) and connect it to Tableau as an extract.	High
2	Data Processing	Use the cleaned file (0 duplicates, 0 missing values); check data types; create Model Year Label and KPI calculated fields; create a Hyper extract.	High
3	Business Analysis	SUM of annual sales by region, model year and battery type; AVG of battery capacity, efficiency, AC charging time, motor power, price and insurance by category, battery type or drivetrain; 6 KPI calculated fields.	High
4	Business Insights & Analytical Outputs	9 worksheets, a dashboard with a KPI strip and 6 charts, and a 3-point Tableau Story answering the three scenarios.	High
5	User Interaction & Analysis	Filters for Market Region, Category and Model Year applied to all sheets; tooltips; sorted bars; story navigation.	Medium
6	Data Update & Reporting	Dashboard and story published on Tableau Public; refreshing the	Medium

		extract updates every chart.	

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	KPIs on top, filters beside them and two rows of three charts – easy to scan.
NFR-2	**Security**	Only public, non-personal market data; dashboard is read-only on Tableau Public.
NFR-3	**Reliability**	Cleaned data (0 duplicates, 0 missing values); KPI values checked against the source data.
NFR-4	**Performance**	Extract of 199,999 rows; charts update within seconds when filters change.
NFR-5	**Availability**	Available 24×7 through the Tableau Public link.
NFR-6	**Scalability**	New data can be added by refreshing the extract; layout supports more charts.